ModelIt!

Student Activity Pack

# When Trees Become Matches

California's Burning Season and the Earth Systems That Fuel It

Grade 5 | NGSS 5-ESS2-1 | Systems and System Models

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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# 1 Pre-Assessment

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**What Do You Know?** Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

**1 What do you already know about wildfires? Have you seen news about California fires?**

**2 What do you think causes wildfires to spread so quickly?**

**3 What is a model? Why do scientists use models?**

**4 What is a system? Can you name some examples of systems?**

# 2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

# 3 Component Analysis

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### What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

### Sort These Components

Rainfall | Wind | Dry Vegetation | Fire Spread

| EXTERNAL | INTERNAL |
|---|---|
| | |
| | |

**Explain your reasoning: Why did you sort each component the way you did?**

# 4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

**? Explain ONE relationship in your model. What happens when one component changes?**

# 5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

## 1 Scenario 1: DROUGHT

Turn RAINFALL to OFF. Leave WIND OFF. Observe what happens to Dry Vegetation and Fire Spread.

MY PREDICTION

WHAT I OBSERVED

## 2 Scenario 2: WINDY DAY

Turn WIND to ON. Leave RAINFALL ON. Observe how wind affects Fire Spread.

MY PREDICTION

WHAT I OBSERVED

## 3 Scenario 3: CALIFORNIA FIRE SEASON

Turn RAINFALL OFF and WIND ON. This simulates real fire season conditions!

MY PREDICTION

WHAT I OBSERVED

# 6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

## Possible New Components

**Temperature** — Hot days dry out plants faster

**Humidity** — Dry air pulls moisture from living plants

**Human Activity** — Campfires, power lines, and construction can spark fires

## My Extension

**A Which new component would you add to the model?**

**B How would it connect to the existing components? (Draw or describe)**

**C What evidence or reasoning supports your connection?**

# 7 Reflection Questions

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### The Big Question

**Why does California burn every single year?** Use evidence from your simulation to support your answers.

**1 How do Earth's atmosphere and biosphere interact during fire season?**

**2 What role does the hydrosphere play in preventing wildfires?**

**3 Why are some years worse for fires than others?**

**4 How do firefighters use their understanding of systems to fight fires?**

**5 What could humans do to reduce wildfire risk?**

# 8 STEM Challenge: Firebreak Engineers!

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## YOUR ENGINEERING MISSION

You learned that DRY VEGETATION is the FUEL for fire spread. Real engineers use FIREBREAKS — areas with no fuel — to stop fires! Now YOU will design firebreaks to protect a community.

### THE SCENARIO

Willow Creek is a small California town surrounded by forest. Fire season is coming! Santa Ana winds blow from the EAST. The town has 2,000 residents, a school, and a hospital. A river runs along the south side. The town has a budget for only 3 firebreaks. Where will you put them?

### Engineering Design Thinking

1. WHAT IS THE PROBLEM YOU NEED TO SOLVE?

2. WHAT ARE THE CONSTRAINTS (BUDGET, TERRAIN, WIND)?

3. WHAT RESOURCES ARE AVAILABLE (RIVER, ROADS)?

4. WHAT TRADE-OFFS MUST YOU CONSIDER?

### My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

**! Explain your design using scientific evidence from the simulation:**

### Real Career Connection

Wildland Firefighters and Fire Scientists protect communities and study fire behavior. They earn $40,000-$90,000/year.